

Name:

Points:

20 /20

VG

Databasutveckling och lagring Final Exam

2025-02-19

Instructions

1. **No computers, no phones, no notes**
2. **No talking with other people in the room**
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Points* box!)
5. For each question, you'll need to answer the question by crossing with an X the right answer.
6. Once you're done:
 - Double-check your answers
 - Hand over your test to me
 - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Point system

There are **16 questions**, each question gives **1 point**, except the last question which gives **5 points**; for a total of maximum **20 points**.

Grading

Points	Grade
0 - 9	IG
10 - 17	G
18 - 20	VG

READ EACH QUESTION CAREFULLY!

1 SQLite and MariaDB are examples of

- 1. NoSQL databases
- ✓ ~~2~~ 2. Relational Databases
- 3. In-memory databases

2 A table is

- ✓ ~~1~~ 1. A collection of structured data
- 2. A technique used to normalize data
- 3. A type of data we can use in our databases

3 A column is

- 1. A technique to normalize data
- ✓ ~~2~~ 2. A field with a certain datatype in a table
- 3. A type of data we can use in our databases

4 A Primary Key

- 1. Is more important than a secondary key
- 2. Is a deprecated technique no longer used in modern RDBMS
- ✓ ~~3~~ 3. Identifies each row in a table uniquely

5 We keep the state of our data consistent at all times by using

- ✓ ~~1~~ 1. Transactions
- 2. Transfusions
- 3. Transitions
- 4. Transmissions
- 5. Translations
- 6. Transformers

6 Given I have a table called users, and I want to set the field active to 0 for all users who have not confirmed their email address (stored in the boolean field email_confirmed), I can run

1. UPDATE * FROM users WHERE active = 0 AND email_confirmed = 0
2. UPDATE users WHERE active = 0 AND email_confirmed = 0
3. UPDATE * FROM users SET email_confirmed = 0 WHERE active = 0
4. UPDATE users SET email_confirmed = 0 WHERE active = 0
5. UPDATE * FROM users SET active = 0 WHERE email_confirmed = 0
- ✓ ~~6.~~ UPDATE users SET active = 0 WHERE email_confirmed = 0

7 Database normalization allow us to

1. Save more data with less storage space
2. Retrieve more data with less memory usage
3. Retrieve more data with less bandwidth usage
- ✓ ~~4.~~ Avoid anomalies
5. Avoid abnormalities

8 1NF states that

- ✓ ~~1.~~ Each field in a table cannot contain multiple values
2. Each table cannot contain multiple fields
3. Each database cannot contain multiple tables
4. Each table must have a primary key

9 To conform to 2NF, a table must conform to 1NF and

1. The primary key must depend on all non-key values
- ✓ ~~2.~~ All non-key values must depend on the entire primary key

10 To conform to 3NF, a table must conform to 1NF

- ✓ ~~1.~~ True
2. False

11 How can I take a backup of a SQLite database?

1. `sqlite3 database.db -backup > backup.sql`
2. `sqlite3 database.db -dump > backup.sql`
3. `sqlite3 database.db .backup > backup.sql`
- ✓ ~~4.~~ `sqlite3 database.db .dump > backup.sql`

12 How can I restore a SQLite backup?

1. `sqlite3 database.db | backup.sql`
- ✓ ~~2.~~ `sqlite3 database.db < backup.sql`
3. `sqlite3 database.db -restore backup.sql`
4. `sqlite3 database.db .restore backup.sql`

13 Indices will speed up all operations on the table

1. True

✓ ~~2.~~ False

14 How many days did Hanukkah of Data consist of?

1. 6
2. 7

✓ ~~3.~~ 8

15 When using SQLite with python, by default INSERT and UPDATE operations need to be manually committed

- ✓ ~~1.~~ True

2. False

16 Explain what a JOIN is (5 points)

✓ A JOIN lets us link separate tables together using primary or foreign keys to retrieve the data we need.